

Supplementary material: Irrigation strategies under contrasting water-storage and rainfall conditions in the North China Plain

Linjia Yao^{1,†}, Bin Chen^{2,†}, Hao Feng², Qiang Yu^{2,*}, Gang Zhao^{2,*}

¹ College of Natural Resources and Environment, Northwest A&F University, Yangling 712100, China

² State Key Laboratory of Soil and Water Conservation and Desertification Control, College of Soil and Water Conservation Science and Engineering, Northwest A&F University, Yangling, Shaanxi 712100, China

† Linjia Yao and Bin Chen contributed equally to this work and share first authorship.

* Corresponding authors: Qiang Yu (yuq@nwfau.edu.cn) and Gang Zhao (gang.zhao@nwfau.edu.cn).

Supplementary methods and results

S1. Observation sources and eligibility

The station archive preserves 132 crop–plot–season trajectories, whole site-year partitions and source metadata. The 7,397 eligible records occur in 127 cases and retain all five observation sites for their available variables.

Calibration and retrospective-evaluation trajectory counts remain 71 and 61. Variable-specific availability and missing combinations are retained in the accompanying case and site tables; missing measurements do not become zero observations.

Wheat LAI above $7 \text{ m}^2 \text{ m}^{-2}$ remains excluded under the study-specific rule. Duplicated six-replicate maize harvest vectors in Yucheng 2012 and 2017 account for 24 excluded grain and biomass records, divided equally between the original calibration and evaluation partitions. These source-based exclusions retain raw values and do not depend on prediction residuals.

Earlier multisite growth estimates retain 480 wheat and 433 maize calibration observations. Current station growth and ET comparisons are conditional on archived management whose full irrigation-log completeness is unresolved. They have zero weight in the water-parameter fit. Wuqiao 2016–2018 supplies 12 documented treatment seasons per crop for that fit (Yang et al., 2024); 2019 supplies four retrospective testing seasons per crop. Seasonal ET, grain and biomass targets refer to the same treatment-season population, while ET contrasts are derived from those seasonal targets.

26 Twenty-six Fengqiu cases retain antecedent moisture initialization from the preceding calendar month. Complete
27 fifteen-layer 0–1.5 m volumetric profiles map conservatively to the represented model layers, preserving depth-
28 integrated storage. The identical eligibility rule applies to 11 original calibration and 15 evaluation cases. Monthly
29 timing, aggregation and subplot support remain unresolved relative to the exact sowing-day state. Original cases
30 remain available when a measured antecedent profile is absent.

31 **Table S1. Crop seasons and calibration/evaluation case counts by site.**

Site	Crop	Years	Calibration cases	Evaluation cases
Fengqiu	Maize	2004, 2005, 2006, 2007, 2008	11	9
Fengqiu	Wheat	2004, 2005, 2006, 2007, 2008	11	10
Gucheng	Maize	2018, 2019, 2020	8	4
Luancheng	Maize	2004, 2005, 2006, 2007, 2008	2	3
Luancheng	Wheat	2004, 2005, 2006, 2007, 2008	2	3
Shangqiu	Maize	2002, 2004, 2005, 2006, 2007	2	6
Shangqiu	Wheat	2001, 2002, 2003, 2004, 2005, 2006, 2007	9	8
Yucheng	Maize	1998, 2000, 2001, 2002, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022	14	9
Yucheng	Wheat	1999, 2000, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022	12	9

32 Years denote archived crop-season groups, which retain all their plot cases. Coverage is intermittent; years absent
33 from the table are not assumed to have observations.

34 **Table S2. Source datasets for evaluated field observations.**

Site	Dataset	DOI	Source
Fengqiu	2004-2008年河南封丘站	10.12199/nesdc.ecodb.mon.	National Ecosystem
	作物生育及生长动态监测	2020.dp2011.fqa.002	Science Data Center /
	数据		Science Data Bank
Fengqiu	2005-2008年河南封丘站	10.12199/nesdc.ecodb.mon.	National Ecosystem
	农田土壤水分物理性质监	2020.dp2011.fqa.006	Science Data Center /
	测数据		Science Data Bank
Gucheng	2018-2021年华北北部不	10.57760/sciencedb.o00119	National Ecosystem
	同播期对玉米生长发育的	.00017	Science Data Center /
	影响数据集		Science Data Bank
Luancheng	2000-2008年河北栾城站	10.12199/nesdc.ecodb.mon.	National Ecosystem
	作物生育及生长动态监测	2020.dp2011.lca.002	Science Data Center /
	数据		Science Data Bank
Shangqiu	2000-2007年河南商丘站	10.12199/nesdc.ecodb.mon.	National Ecosystem
	作物生育及生长动态监测	2020.dp2011.sqa.003	Science Data Center /
	数据		Science Data Bank
Yucheng	1998-2006年山东禹城站	10.12199/nesdc.ecodb.mon.	National Ecosystem
	作物生育及生长动态监测	2020.dp2011.yca.003	Science Data Center /
	数据		Science Data Bank
Yucheng	禹城站2005-2022年生物	10.12199/nesdc.ecodb.2021	National Ecosystem
	要素观测数据集	YFF0703900.yca.2025.2	Science Data Center /
			Science Data Bank
Yucheng	禹城站2005-2022年水环	10.12199/nesdc.ecodb.2021	National Ecosystem
	境要素观测数据集	YFF0703900.yca.2025.3	Science Data Center /
			Science Data Bank

35 Dataset DOIs identify official source records; original metadata and observation-row provenance are retained in the
36 reproducibility archive.

37 **S2. Crop processes, parameter definitions and water accounting**

38 **S2.1. Model structure and phenological development**

Open Crop Model simulates phenology, green leaf area, live and dead aboveground biomass, grain dry matter, rooting depth and layered soil-water storage. Inputs comprise weather, soil profiles, crop parameters, dated sowing and harvest, and irrigation events. Soil water carries continuously between crop and fallow segments in regional histories; Wuqiao crop starts retain their separately specified storage conditioning.

Wheat stage development combines temperature, vernalization and photoperiod modifiers with inherited stage requirements. Maize uses a subdaily temperature response reconstructed from daily extrema, with base, optimum and maximum temperatures of 6, 30 and 44 °C. The shared medium-maturity maize card requires approximately 1,909 °C d to reach R6. DVS anchors are wheat BBCH 09, 61 and 89 and maize VE, R1 and R6 (Meier, 2018).

Wheat dough stage and full ripeness retain their distinct definitions.

Seed-layer water above wilting-point storage permits germination. The represented layer is selected by sowing depth, with a 50-mm prior when no depth is supplied. Thermal development and wheat vernalization begin from the first eligible germination day rather than accumulating during dry-seed days. This layer-average availability is an effective establishment condition, not a measured seed-scale water potential. Fixed stage coefficients therefore do not imply fixed calendar-stage predictions across water states. New biomass production ends at wheat full ripeness or maize physiological maturity; green maize area can persist under the terminal senescence rate until harvest. Harvest is a separate management endpoint and does not substitute for an observed maturity date.

S2.2. Canopy development, biomass accumulation and grain yield

Green cover is $f_c = 1 - \exp(-k_L \text{LAI})$. Daily aboveground dry-matter production is $\Delta B = 10R_s p_{\text{PAR}} f_c \varepsilon f_T f_W^y f_N$, where R_s is incident shortwave radiation, $p_{\text{PAR}} = 0.45$, ε is RUE per MJ of intercepted PAR, f_T is the temperature modifier, f_N is the prescribed nutrition factor and f_W is realized transpiration divided by remaining potential transpiration demand after wet-canopy evaporation. The factor 10 converts g m^{-2} to kg ha^{-1} . PAR enters once (Kiniry et al., 1989). The fitted empirical assimilation exponents are 1.85985 for wheat and 0.25000 for maize. They govern carbon response separately from leaf-expansion and leaf-survival sensitivity.

SLA is linearly interpolated through DVS 0, 0.5, 1 and 2, with the DVS-2 value equal to DVS 1. New leaf dry matter acquires area using current SLA; established cohorts retain their birth area-to-mass ratio (de Wit and Berghuijs, n.d.). Thermal senescence hazard $h(\text{DVS})$ is interpolated through DVS 0, 1, 1.4, 1.7 and 2, with a constant preflowering rate and increasing grain-filling rates (Tan et al., 2020). Daily survival

is $\exp[-h(\text{DVS}) \max(T_{\text{mean}} - T_{\text{base}}, 0)](1 - q_s)$, with $q_s = 0.025(1 - f_w f_N)$. The terminal hazard continues at maize R6, whereas wheat BBCH 89 senesces remaining green leaves. LAI emerges from the cohort area balance; no maximum-LAI parameter or cap enters this formulation.

Leaf allocation follows crop development, with separate early and late vegetative multipliers for maize and additional temperature, water-supply and nutrition limits on leaf expansion. Initial LAI is an emergence condition. Stage-specific SLA and senescence coefficients retain their inherited estimates and conditional trait priors (APSIM Initiative, 2026; de Wit, 2022; de Wit, 2020); their values do not represent universal physiological bounds.

Grain number is fixed at the first grain-set day as $N_g = 0.1B_f c_g$, where B_f is simulated aboveground dry mass immediately before grain set and c_g is the grain-number coefficient. Grain set starts at wheat BBCH 61 and maize R1. Maximum grain dry mass is $Q_g = 10N_g m_g$, with fixed m_g of $0.045 \text{ g grain}^{-1}$ for wheat and $0.300 \text{ g grain}^{-1}$ for maize (APSIM Initiative, 2026). Assimilate supply and development-dependent filling capacity constrain grain accumulation. Wheat remobilization transfers an explicit subset of stem mass to grain while conserving total aboveground mass. Live and dead leaves, stems and grain remain in the aboveground carbon pool; observed yield or harvest index does not initialize grain.

Prescribed nutrition factors distinguish archived fertilized and unfertilized management classes. Wuqiao and regional simulations retain the fertilized class. These factors describe effective growth limitations; the model does not resolve a mechanistic nitrogen balance.

S2.3. Soil-water balance and actual evapotranspiration

Potential transpiration and exposed-soil evaporation are $T_p = K_T \text{ET}_0 f_c$ and $E_p = K_E \text{ET}_0 (1 - f_c)$, with ET_0 supplied by AgERA5. Rainfall interception fills a canopy store of capacity $1 \text{ mm} \times f_c$. Wet-canopy evaporation reduces the transpiration demand remaining within each outer water step. Actual ET is $T_a + E_s + E_c$, the sum of transpiration, soil evaporation and canopy evaporation. Wuqiao residue cover is unmeasured and its exposed-soil boundary is retained; prescribed organic mulch, where supplied, reduces potential soil evaporation using the FAO-56 cover approximation (Allen et al., 1998).

Two-stage evaporation depends on actual surface-layer storage. Total evaporable water is

$\text{TEW} = (\theta_{\text{FC}} - \theta_{\text{AD}})\Delta z_1$ and readily evaporable water is $\text{REW} = 0.30\text{TEW}$. Surface depletion is

$D_e = (\theta_{\text{FC}} - \theta_1)\Delta z_1$. The evaporation reduction is $K_r = \min \left[1, \max \left(0, \frac{\text{TEW} - D_e}{\text{TEW} - \text{REW}} \right) \right]$, with actual

94 extraction bounded by storage above air dry. Ponded water evaporates before soil storage. This storage
 95 closure follows the two-stage drying principle (Allen et al., 1998) without resolving a complete dual-crop-
 96 coefficient surface balance.

97 Root activity integrates an exponential depth distribution over rooted parts of each layer and is normalized
 98 over the represented rooted volume. Local depletion stress is $s_i = \min \left[1, \frac{\max(0, W_i - \theta_{WP,i} \Delta z_i)}{(1-p)(\theta_{FC,i} - \theta_{WP,i}) \Delta z_i} \right]$, where p
 99 is the fitted readily available water fraction. Base uptake combines root activity, local stress, atmospheric
 100 demand and the daily extraction coefficient; accessible storage and extraction capacity bound each
 101 contribution. The compensation fraction allocates a bounded part of unmet demand to rooted layers whose
 102 stress factors equal or exceed the activity-weighted mean. Total uptake remains bounded by demand times
 103 the best rooted-layer stress factor and by available extraction capacity. Compensation redistributes
 104 demand and does not eliminate uniformly dry-profile stress.

105 Effective saturation is $S_e = \frac{\theta - \theta_{AD}}{\theta_{SAT} - \theta_{AD}}$. With $m = 1 - \frac{1}{n}$, the retention relation is $S_e = \left[1 + (\alpha | h |)^n \right]^{-m}$
 106 (van Genuchten, 1980), and conductivity is $K = K_{sat} S_e^{1/2} \left[1 - (1 - S_e^{1/m})^m \right]^2$ (Mualem, 1976; van
 107 Genuchten, 1980). Adjacent half-layer saturated resistances define $K_{sat, face} = \frac{d}{\frac{\Delta z_i}{2K_{sat,i}} + \frac{\Delta z_{i+1}}{2K_{sat,i+1}}}$, where
 108 $d = \frac{\Delta z_i + \Delta z_{i+1}}{2}$. Relative-arithmetic face mobility is $r_{face} = \frac{\Delta z_{i+1} r_i + \Delta z_i r_{i+1}}{\Delta z_i + \Delta z_{i+1}}$, with $r_i = \frac{K_i}{K_{sat,i}}$. Thus
 109 $K_{face} = K_{sat, face} r_{face}$. This interface approximation permits a wet layer to supply a dry neighbor while
 110 retaining the resistance of low-conductivity material.

111 Signed interface flux is $q_i = K_{face} \left[1 + \frac{h_i - h_{i+1}}{d} \right]$, positive downward. Transfers have equal and opposite
 112 contributions in neighboring layer balances and are bounded by donor storage and receiving pore capacity.
 113 Twenty-four outer steps partition daily forcing, evaporation and uptake; adaptive internal transport steps
 114 limit net θ increments to 0.005. The free-drainage bottom flux equals the current unsaturated conductivity
 115 under unit hydraulic gradient. Upward redistribution is an internal profile flux, with no external capillary-
 116 rise supply. This bounded explicit layered approximation does not resolve the full implicit Richards
 117 system or saturated positive pressure.

118 For a field of profile storage S , the accounting identity is $\Delta S = P + I - ET_a - D - Q$. Applied irrigation I
 119 is a field-boundary input. Flood-irrigation scenarios wet the entire field, with application, drift and
 120 conveyance losses set to zero at that boundary. Drainage D leaves the modeled profile and is distinct from

121 water reaching an aquifer after deeper vadose-zone transport (Wu et al., 2023).

122 **S2.4. Hydraulic derivation, shared crop parameters and numerical checks**

123 Saxton–Rawls equations use normalized sand and clay fractions and organic matter percent (Saxton and
124 Rawls, 2006). SoilGrids carbon in dg kg^{-1} is converted to organic matter percent by $\frac{1.724 \times \text{SOC}}{100}$; bulk
125 density in cg cm^{-3} is divided by 100; coarse fragments in permille are divided by 1,000 (ISRIC — World
126 Soil Information, 2020). Regional porosity uses $1 - \frac{\text{BD}}{2.65}$, with prescribed density and gravel adjustments.
127 Air-dry water content retains its declared prior. Field-capacity, wilting-point and saturation ordering is
128 checked before simulation; unavailable complete profiles remain excluded under the spatial-source rule.

129 The van Genuchten α and n values are conditional hydraulic priors reconstructed from supplied field-
130 capacity and wilting-point endpoints at pressure heads of $-3,300$ and $-150,000$ mm, with θ_{AD} as residual
131 water and θ_{SAT} as saturated content. Endpoint reconstruction preserves those storage thresholds rather
132 than fitting curves to crop outcomes. Neither depth-resolved Wuqiao retention measurements nor exact
133 seasonal initial layer distributions are available. The separate Wuqiao particle-density prior is 2.75 g cm^{-3} .

134 Table S3 contains effective shared crop values after applying the frozen fit multipliers. Regional and Wuqiao inputs
135 use identical shared cards within each crop. Maximum root depth is limited by the supplied profile depth. Stage-
136 specific SLA and senescence, fixed grain mass and inherited growth coefficients remain distinct from the eight
137 water coefficients fitted in the current run. Near-zero root-density decay yields approximately uniform activity over
138 represented rooted volume; it does not identify an observed root-mass profile.

139 The 164 selected station and field trajectories conserve daily water and aboveground carbon, with maximum
140 absolute residuals of 1.00×10^{-12} mm and 6.71×10^{-12} kg ha $^{-1}$. Exact inputs, observation identities, frozen
141 parameters, source snapshots and SHA-256 provenance accompany the predictions. Conservation establishes
142 arithmetic consistency; hydraulic and crop-response accuracy depend on the conditional observations and
143 retrospective field test.

144 Eight existing water-response coefficients per crop are estimated from documented Wuqiao management in 2016–
145 2018: the transpiration multiplier, soil evaporation coefficient, readily available water fraction, assimilation water-
146 stress exponent, root-density decay, maximum rooting depth, $\log_{10}$ daily root extraction coefficient and root
147 compensation fraction. RUE, leaf allocation and grain-number multipliers remain fixed. Each crop contributes 12
148 seasonal ET, 12 dry-grain and 12 dry-biomass targets, together with three W3–W0 ET contrasts derived from the

149 same seasonal targets. The criterion retains weights of 0.325, 0.175, 0.15 and 0.10 for those components. ET
150 residual scales are $\max(50 \text{ mm}, 0.15 \times \text{observed ET})$; grain and biomass scales are $\max(1,000 \text{ kg ha}^{-1}, 0.20 \times$
151 $\text{observed dry mass})$. Contrast residuals use 50 mm. Eight coefficient-prior terms have total weight 0.01, with
152 inherited bounds, centres and scales.

153 Bounded least squares starts from the inherited water vector, with coefficients scaled to their prescribed intervals, a
154 40-evaluation limit and convergence tolerances of 8×10^{-5} . Reference and refitted coefficients use the identical
155 field-only criterion; the equal-crop mean calibration data loss is 0.592580 for the reference and 0.577934 for the
156 selected refit. Selection excludes prior penalties from this comparison and precedes 2019 testing. Station
157 observations, daily ET and phase ET with unresolved sampling windows have zero weight in the water-parameter
158 fit. Source observations, weather, soil priors, initialization rules and native process equations remain fixed during
159 this refit.

160 **Table S3. Shared crop parameters transferred to regional simulations.**

Parameter	Unit	Wheat	Maize
Radiation-use efficiency	g MJ ⁻¹ intercepted PAR	3.082080	3.379974
Specific leaf area at DVS 0	m ² g ⁻¹	0.028482	0.024045
Specific leaf area at DVS 0.5	m ² g ⁻¹	0.027578	0.029960
Specific leaf area at DVS 1	m ² g ⁻¹	0.023462	0.024000
Leaf senescence rate before flowering	(°C d) ⁻¹	0.000536	0.000010
Leaf senescence rate at DVS 1.4	(°C d) ⁻¹	0.001907	0.000023
Leaf senescence rate at DVS 1.7	(°C d) ⁻¹	0.005327	0.001650
Leaf area index at emergence	m ² m ⁻²	0.141900	0.070844
Grain number per biomass at grain set	grains g ⁻¹	26.272389	4.242740
Maximum individual grain mass	g grain ⁻¹	0.045000	0.300000
Transpiration coefficient	dimensionless	0.838813	1.023273
Soil evaporation coefficient	dimensionless	1.295161	0.100000
Root water uptake coefficient	d ⁻¹	0.042671	0.033543
Leaf biomass partitioning multiplier	dimensionless	1.010797	—
Early vegetative leaf partitioning multiplier	dimensionless	—	0.912989
Late vegetative leaf partitioning multiplier	dimensionless	—	1.779539
Stem reserve fraction	dimensionless	0.270612	—
Stem reserve release fraction	dimensionless	0.067334	—
Assimilation water-stress exponent	dimensionless	1.859851	0.250000
Readily available water fraction	dimensionless	0.543149	0.674994
Root density decay	m ⁻¹	3.41 × 10 ⁻¹⁰	2.27 × 10 ⁻¹³
Maximum rooting depth	mm	1888.654910	1699.999998
Root compensation fraction	dimensionless	1.000000	6.62 × 10 ⁻⁸
Readily evaporable water fraction	dimensionless	0.300000	0.300000

161 Effective shared crop parameters from the selected current cards. RUE uses intercepted PAR with PAR fraction
162 0.45. DVS-2 SLA equals its DVS-1 value; senescence is a thermal hazard with a separate daily stress loss. Wheat
163 BBCH 89 terminates remaining green leaf area; maize retains its terminal hazard at R6. Grain mass is fixed. Water

164 coefficients are estimated from Wuqiao 2016–2018; growth coefficients retain inherited multisite estimates. Root
165 depth is limited by profile depth. Dashes denote crop-specific processes. No maximum-LAI parameter is fitted.

166 **S3. Study-area and spatial aggregation checks**

167 The SRTM-derived terrain grid (NASA JPL, 2013) contains 720 eligible lowland components; the connected
168 component retained by the operational definition covers 404,658 km². Disconnected eligible terrain totals 6,465 km²
169 and contains approximately 7,068 ha of CPM-corrected rotation area. The main definition reaches the southern and
170 eastern bounding-box edges, so geographic sensitivity remains relevant to regional interpretation. Nine elevation–
171 slope alternatives combine elevations 100, 200 and 300 m with slopes 1°, 2° and 3°. Seven annual mapped-area
172 layers cover 2015–2021.

173 A common-extent comparison between ChinaCP plus CPM (Qiu et al., 2022) and the 2020 China Crop Dataset
174 layer (Fu et al., 2026; Fu et al., 2025) yields 9.694 and 7.841 million ha, respectively; CCD2020 is 19.11% lower on
175 that common support. This disagreement is map uncertainty rather than proof of either product’s accuracy. The
176 modeled 2020 area is held fixed to isolate climate and irrigation effects. Historical crop expansion, rotation
177 switching and double-counting of harvested crop area are not estimated.

178 Thirty-two actual-cell representatives use rotation-area weights and selection-period climate means. Eight
179 purposively selected non-representative cells test approximation under all five continuous irrigation fractions. Each
180 crop–fraction comparison retains 96 evaluation cell-years in 2014–2025. Absolute grain, ET and management-
181 response differences compare actual-cell with representative-cell model predictions. The sample measures
182 aggregation sensitivity rather than observed regional accuracy or a probabilistic error bound (Table S4; Figure S1).

183 **Table S4. Crop-model spatial aggregation errors in eight purposively selected simulation cells.**

Crop	Irrigation fraction	Variable	Unit	N	RMSE	Bias
Maize	0.000	Grain yield	t ha ⁻¹	96	1.17	0.18
Maize	0.000	ET	mm	96	35.21	6.41
Maize	0.250	Grain yield	t ha ⁻¹	96	0.72	0.13
Maize	0.250	ET	mm	96	27.39	5.17
Maize	0.500	Grain yield	t ha ⁻¹	96	0.50	0.08
Maize	0.500	ET	mm	96	22.19	3.82
Maize	0.750	Grain yield	t ha ⁻¹	96	0.39	0.08
Maize	0.750	ET	mm	96	17.85	4.38
Maize	1.000	Grain yield	t ha ⁻¹	96	0.35	0.09
Maize	1.000	ET	mm	96	14.97	5.69
Wheat	0.000	Grain yield	t ha ⁻¹	96	1.49	0.19
Wheat	0.000	ET	mm	96	51.66	-3.18
Wheat	0.250	Grain yield	t ha ⁻¹	96	1.39	0.07
Wheat	0.250	ET	mm	96	43.32	-4.26
Wheat	0.500	Grain yield	t ha ⁻¹	96	1.19	-7.41 × 10 ⁻³
Wheat	0.500	ET	mm	96	37.97	-4.56
Wheat	0.750	Grain yield	t ha ⁻¹	96	0.89	-0.01
Wheat	0.750	ET	mm	96	31.90	-2.05
Wheat	1.000	Grain yield	t ha ⁻¹	96	0.64	0.11
Wheat	1.000	ET	mm	96	24.70	3.21

184 Current model aggregation discrepancies against eight actual-cell simulations under all five irrigation fractions.

185 Grain is converted from kg ha⁻¹ to t ha⁻¹ in every displayed grain-valued column. Exactly zero irrigation

186 discrepancies are omitted from display; full inputs and metrics remain in the current-results product.

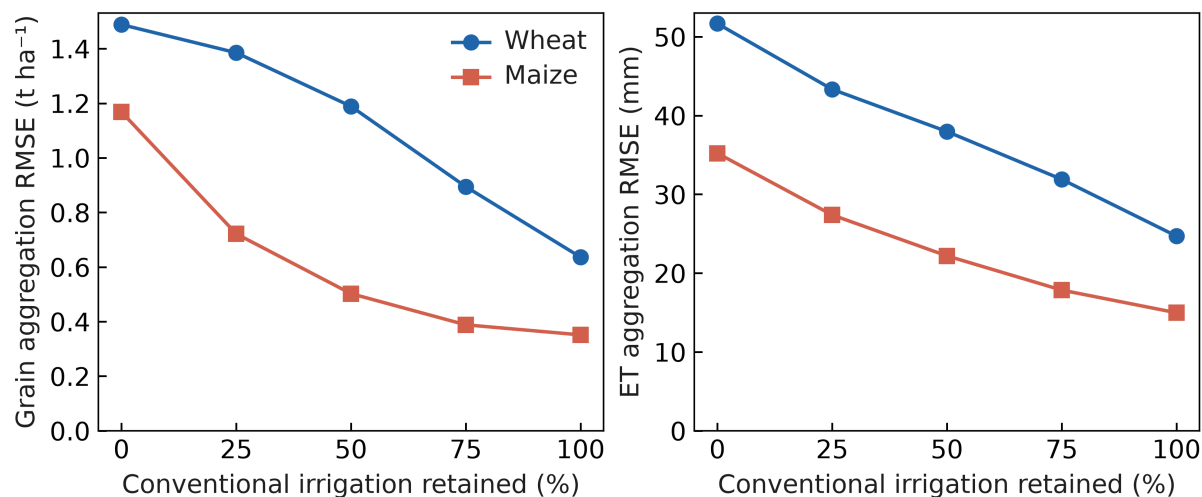


Figure S1. Current crop-model spatial aggregation errors under all five irrigation fractions in eight purposively selected cells during 2014–2025. Actual-cell and representative predictions share the current native model and crop parameters. These discrepancies assess spatial representation rather than field-observation accuracy.

S4. Annual field yields and irrigation-treatment contrasts

Annual Wuqiao grain targets comprise sixteen treatment-year means per crop in 2016–2019, digitized with standard errors from the primary Supplementary Figure S2. Grain retains its 13% moisture basis in comparisons. The twelve 2016–2018 means per crop enter calibration after a single multiplication by 0.87 to dry matter. The four 2019 means remain outside fitting and selection but had been inspected during earlier development. Four-year means summarize the same years and supply no additional independent targets.

Annual grain calibration RMSE is 1.249 t ha⁻¹ for wheat and 2.267 t ha⁻¹ for maize; retrospective-testing RMSE is 3.161 and 1.700 t ha⁻¹ at 13% moisture. Within-year W1–W3 minus W0 grain-contrast RMSE is 0.519 and 1.669 t ha⁻¹ in calibration and 1.316 and 1.323 t ha⁻¹ in 2019 (Table S5; Figure S2). Maize treatment labels retain preceding wheat management because maize irrigation is identical within each year.

All printed seasonal ET targets remain eligible after matching the 2016 maize rainfall period. Maize-season rainfall of 499.1 mm excludes the 100.5 mm during wheat from the 599.6-mm rotation total. Reconstructed balance residuals for W0–W3 are +0.1, +0.1, -0.9 and +0.1 mm. Source ET and depletion values remain unchanged. Storage-depletion diagnostics share the source ET balance and its unconfirmed exact sampling dates; they are conditional accounting comparisons rather than independent soil-water validation.

205 **Table S5. Wuqiao annual grain and within-year treatment contrasts by fitting and testing partition.**

Crop	Dataset	Comparison	N	RMSE (t/ha)	Bias (t/ha)	NSE	R ²
Maize	Calibration (2016–2018)	Annual grain yield	12	2.267	-1.738	-2.940	0.128
Maize	Calibration (2016–2018)	Within-year W1–W3 minus W0	9	1.669	0.296	-0.788	0.076
Maize	Retrospective evaluation (2019)	Annual grain yield	4	1.700	1.403	-3.641	0.972
Maize	Retrospective evaluation (2019)	Within-year W1–W3 minus W0	3	1.323	-0.836	-1.403	1.000
Wheat	Calibration (2016–2018)	Annual grain yield	12	1.249	-0.445	0.746	0.817
Wheat	Calibration (2016–2018)	Within-year W1–W3 minus W0	9	0.519	0.064	0.857	0.882
Wheat	Retrospective evaluation (2019)	Annual grain yield	4	3.161	3.108	-4.654	0.870
Wheat	Retrospective evaluation (2019)	Within-year W1–W3 minus W0	3	1.316	-1.312	-0.712	0.994

206 Grain is at 13% moisture. Calibration comprises 2016–2018 and retrospective testing comprises 2019. Source
207 means and standard errors are digitized from Supplementary Figure S2, with pixel tolerances near 0.043 t ha⁻¹ for
208 wheat and 0.056 for maize. Contrasts compare W1–W3 with same-year W0. Four-year means overlap these annual
209 targets; maize contrasts retain preceding wheat histories.

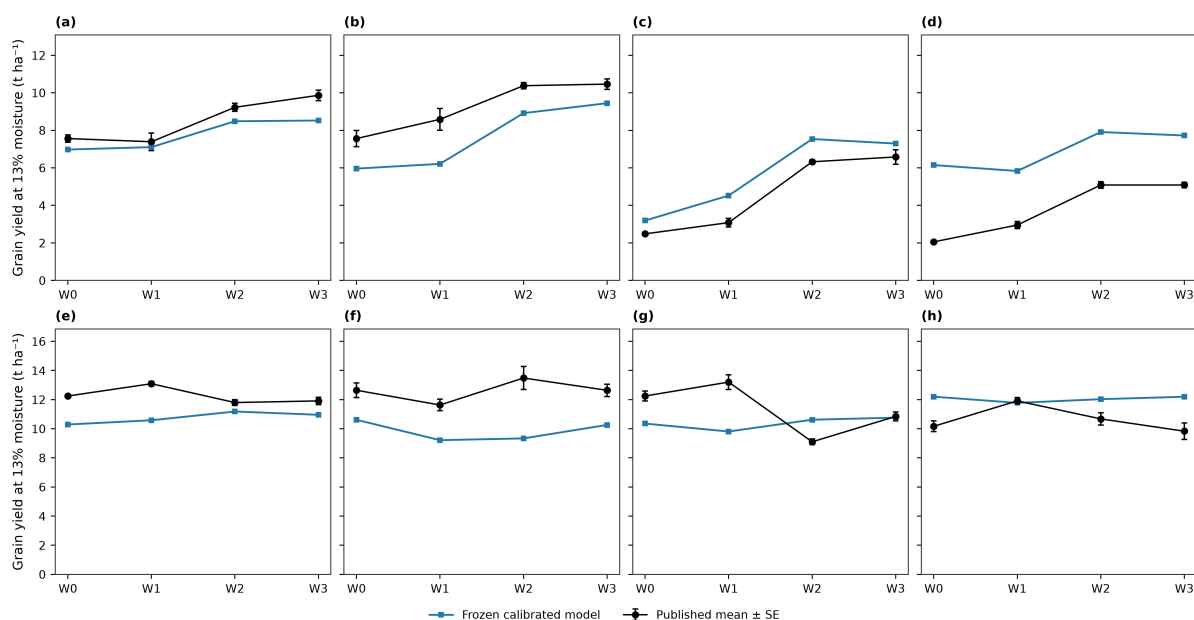


Figure S2. Annual Wuqiao grain at 13% moisture under the selected current parameters. Panels (a–d) show wheat and (e–h) maize; columns from left to right show 2016, 2017, 2018 and 2019. Black points and error bars are published means $\pm$ standard error of three replicates; blue squares show current predictions. The 2016–2018 treatment means contribute to water calibration after conversion to dry grain; 2019 is retrospective testing, excluded from fitting and selection. Maize irrigation is identical within each year and labels retain preceding wheat management. Source means and SE are digitized from Supplementary Figure S2 of the primary experiment.

S5. Class exposure, continuous states and rainfall-only controls

All nine common GRACE-available years occupy lower-storage classes; three other years use conventional fallback. The selected current quotas are identical across storage strata within each rainfall group. Matched rainfall-only and storage–rainfall histories have identical actions, crop states and outcomes throughout the modeled record. Available-rainfall controls apply reduced quotas during monitoring gaps and therefore create different carried water states before later common years. These history effects are separate from the value of the storage classification. Table S7 reports all-year and common-year retention and water use; Table S8 reports the matched and available-rainfall controls.

224 **Table S6. Class exposure and missing-storage coverage by period.**

Period	Class	Unit-years	Years	Area-years (ha yr)	Area-year fraction
evaluation	Lower storage / lower rainfall	174	9	55410278.787	0.468
evaluation	Lower storage / higher rainfall	114	9	33389136.355	0.282
evaluation	Missing storage	96	3	29599805.048	0.250
selection	Higher storage / lower rainfall	68	6	19466918.675	0.179
selection	Higher storage / higher rainfall	124	6	39732691.420	0.366
selection	Lower storage / lower rainfall	109	5	34747673.179	0.320
selection	Lower storage / higher rainfall	51	5	14585335.233	0.134

225 **Table S7. Continuous class-adaptive irrigation evaluation during 2014–2025.**

Strategy	Period	Years	Grain retained (%)	Irrigation cut (mm)	ET cut (mm)	Minimum grain (%)
Storage– rainfall 95%	Nine GRACE years	9	96.270	130.721	36.842	92.115
Storage– rainfall 95%	All twelve years	12	96.818	98.040	29.733	92.115
Storage– rainfall 98%	Nine GRACE years	9	99.626	35.721	6.171	98.579
Storage– rainfall 98%	All twelve years	12	99.622	26.790	5.208	98.579
Conventional	Nine GRACE years	9	100.000	0.000	0.000	100.000
Conventional	All twelve years	12	100.000	0.000	0.000	100.000

226 Current continuous histories. Grain retention is a ratio of period-mean regional dry-grain production to conventional
 227 irrigation. ET includes crops and fallows. All-year summaries include conventional fallback; common-year
 228 summaries use the nine GRACE-available years.

229 **Table S8. Continuous GRACE–rainfall and rainfall-only strategy comparison during 2014–2025.**

Strategy	Period	Grain retained (%)	Irrigation (mm)	ET (mm)
Storage–rainfall 95%	Nine GRACE years	96.27	249.279	848.527
Storage–rainfall 95%	All twelve years	96.82	281.960	857.698
Storage–rainfall 98%	Nine GRACE years	99.63	344.279	879.197
Storage–rainfall 98%	All twelve years	99.62	353.210	882.223
Rainfall 95%, available	Nine GRACE years	95.40	249.279	844.203
Rainfall 95%, available	All twelve years	95.04	248.565	842.945
Rainfall 95%, matched	Nine GRACE years	96.27	249.279	848.527
Rainfall 95%, matched	All twelve years	96.82	281.960	857.698
Rainfall 98%, available	Nine GRACE years	99.41	344.279	878.151
Rainfall 98%, available	All twelve years	99.36	343.565	879.669
Rainfall 98%, matched	Nine GRACE years	99.63	344.279	879.197
Rainfall 98%, matched	All twelve years	99.62	353.210	882.223

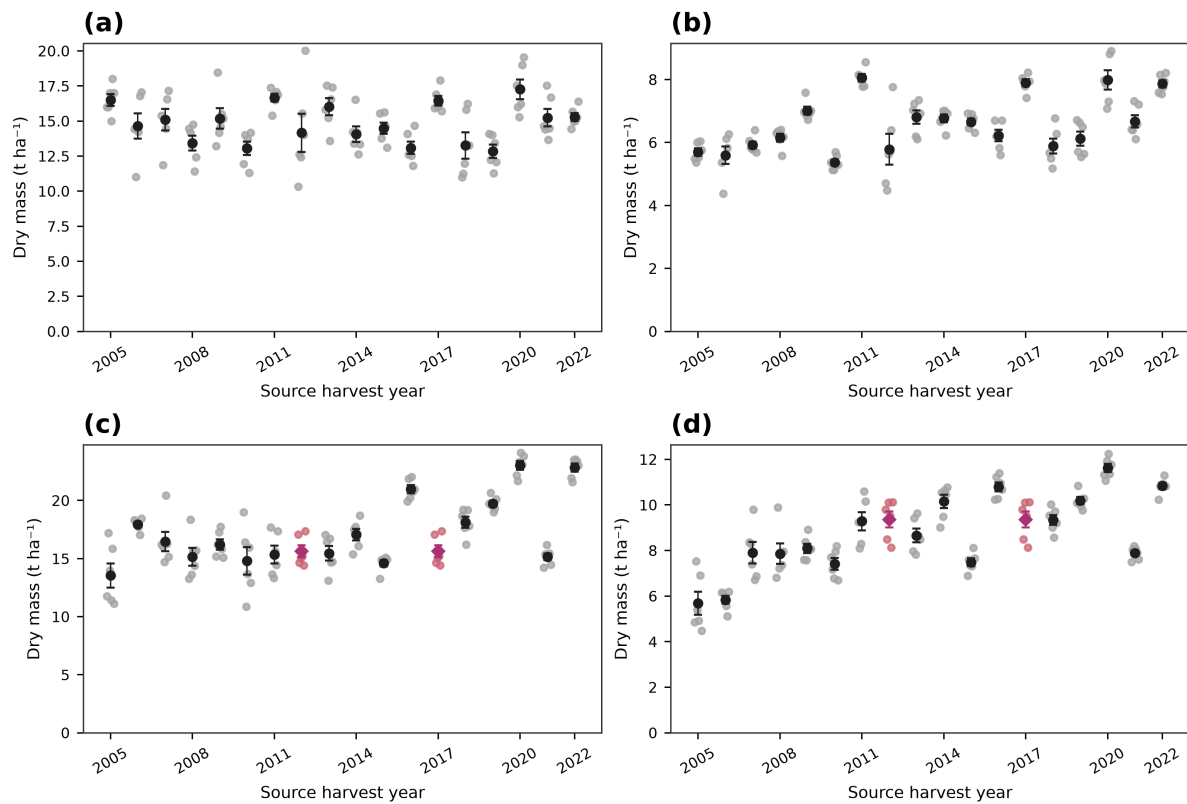
230 Matched controls share the complete GRACE-availability and fallback calendar. Available-rainfall controls act
 231 during missing-storage years and alter subsequent soil states. All values use current parameters; differences do not
 232 identify causal groundwater responses.



233 **Figure S3. Simulated 0–2 m soil-water storage in representative units (a) 0 and (b) 31. State carries continuously through wheat,**
 234 **maize, fallows and annual irrigation choices. Displayed histories compare conventional irrigation with selected current strategies;**
 235 **they are model states rather than observed groundwater levels.**

236 **S6. Harvest data quality and crop model recalibration**

237 Original Yucheng harvest sources contain identical six-replicate maize vectors in 2012 and 2017 under different
 238 date and cultivar labels. Both groups are retained as raw evidence and excluded from scored grain and harvest
 239 biomass. Eligibility depends on source authenticity rather than model residuals. Current plots preserve observed
 240 quadrat variation and the original whole site-year partitions (Figure S4).



241 **Figure S4. Original Yucheng harvest observations at YCAZH01ABC_01. Panels (a, b) show wheat aboveground biomass and grain;**
 242 **panels (c, d) show maize aboveground biomass and grain. Faint points are six quadrat observations; dark markers and error bars**
 243 **are means $\pm$ replicate standard error. Magenta diamonds identify the identical full maize source vectors in 2012 and 2017. Raw**
 244 **values remain displayed for source inspection; both groups are excluded from comparisons with quality-controlled observations.**
 245 **Error bars describe sampling variation, rather than model or policy uncertainty.**

246 **S7. Spatial decomposition under the common irrigation budget**

247 Frozen area shares weight complete current irrigation histories before spatial aggregation. Dry-grain differences in
 248 kg ha⁻¹ multiply represented hectares and divide by 1,000 for tonnes; water depths multiply hectares by 10 for cubic
 249 metres. Independent sums reproduce regional contrasts. Latitude bands describe the distribution of gains and losses,
 250 while the 3,641 mapped cells inherit 32 simulated representative responses (Table S9).

251 **Table S9. Spatial decomposition at a common 50% regional irrigation reduction during 2014–2025.**

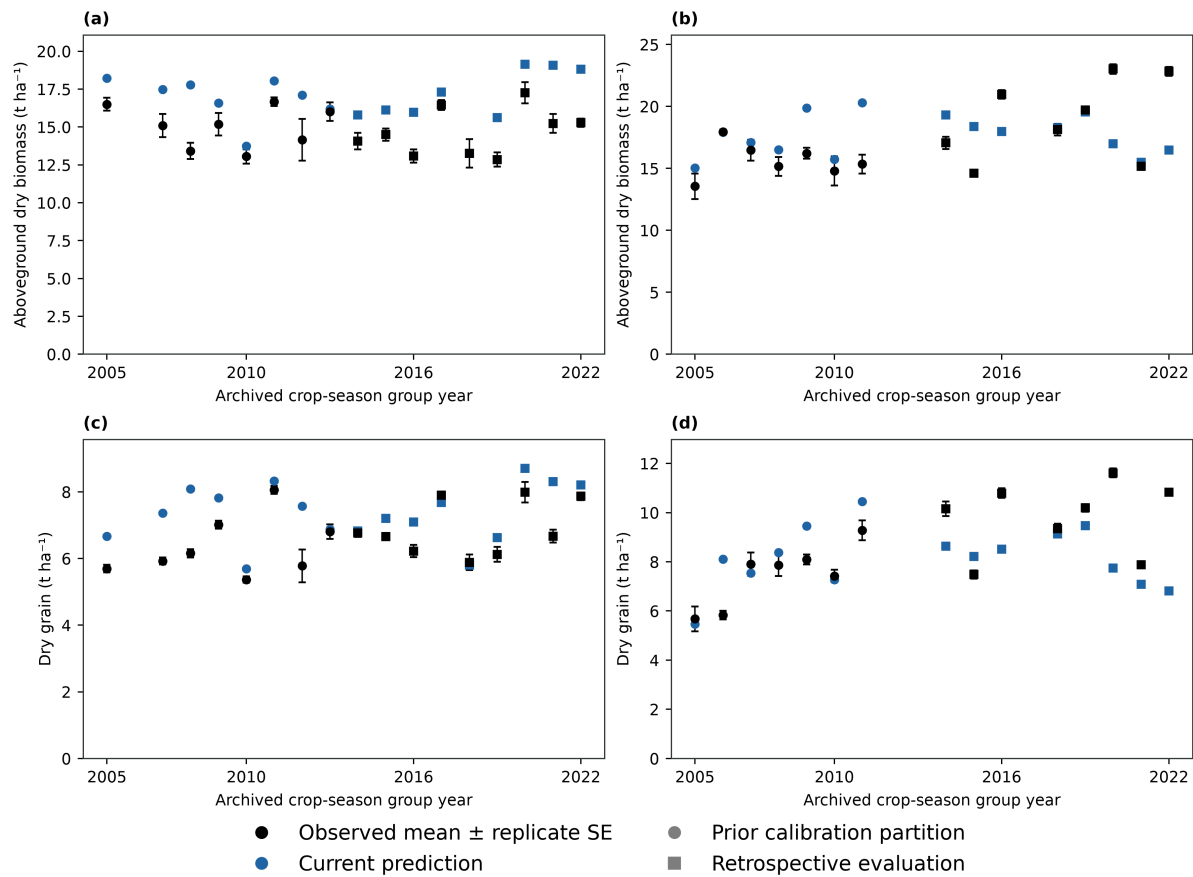
Latitude band	Area share (%)	Targeted irrigation fraction	Grain change (t ha ⁻¹)	ET change (mm)	Grain contribution (Mt)
Southern (32–36°N)	60.733	0.305	-0.357	-25.599	-2.142
Central (36–38°N)	31.369	0.799	2.415	93.497	7.475
Northern (38–41°N)	7.897	0.813	2.420	88.358	1.886

252 Current targeted-minus-uniform means at the same 50% regional irrigation cut. Latitude bands are reporting
253 partitions. Source cells inherit representative responses and are not independent observations.

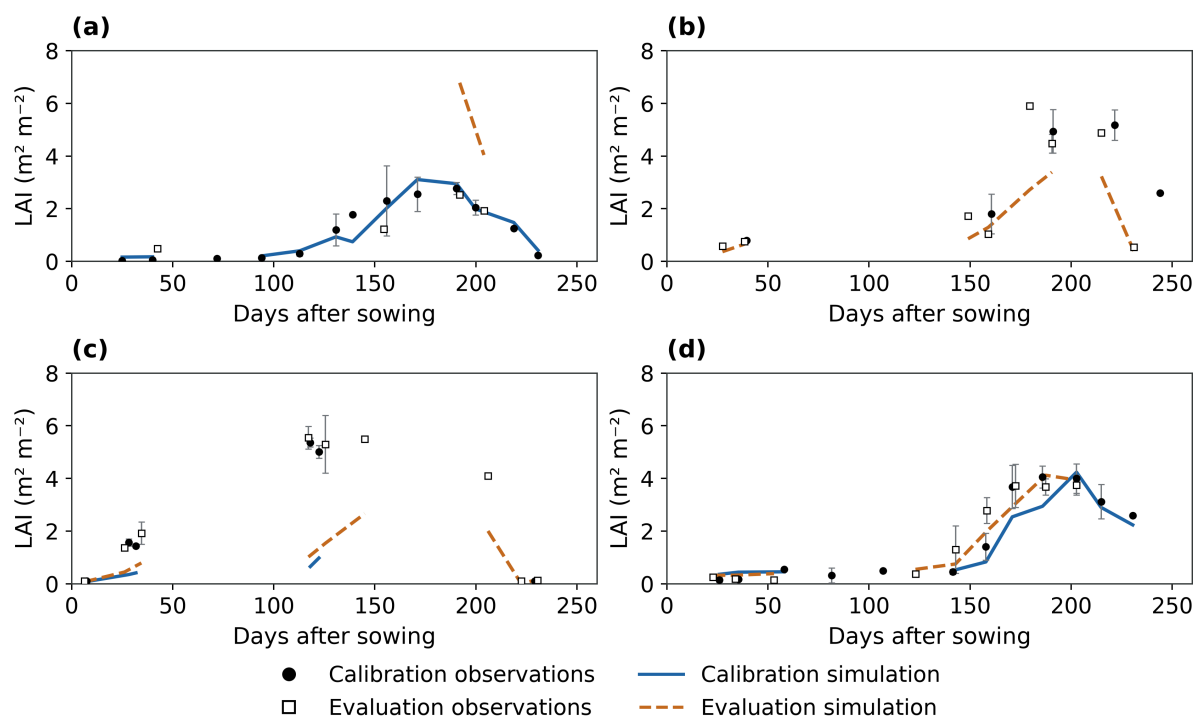
254 **S8. Seasonal crop trajectories and harvest outcomes**

255 Current seasonal curves use the selected frozen parameters and all sites with eligible observations for each variable.
256 Observation and prediction channels share windows, case identities and aggregation weights. Bins span 15 days for
257 wheat and 7 days for maize; site-year contributions and populations can change among bins. Standard errors
258 describe variation across site-year means and are omitted for single-year bins. Missing bins remain gaps. Unbinned
259 records determine performance scores (Figures S6–S11).

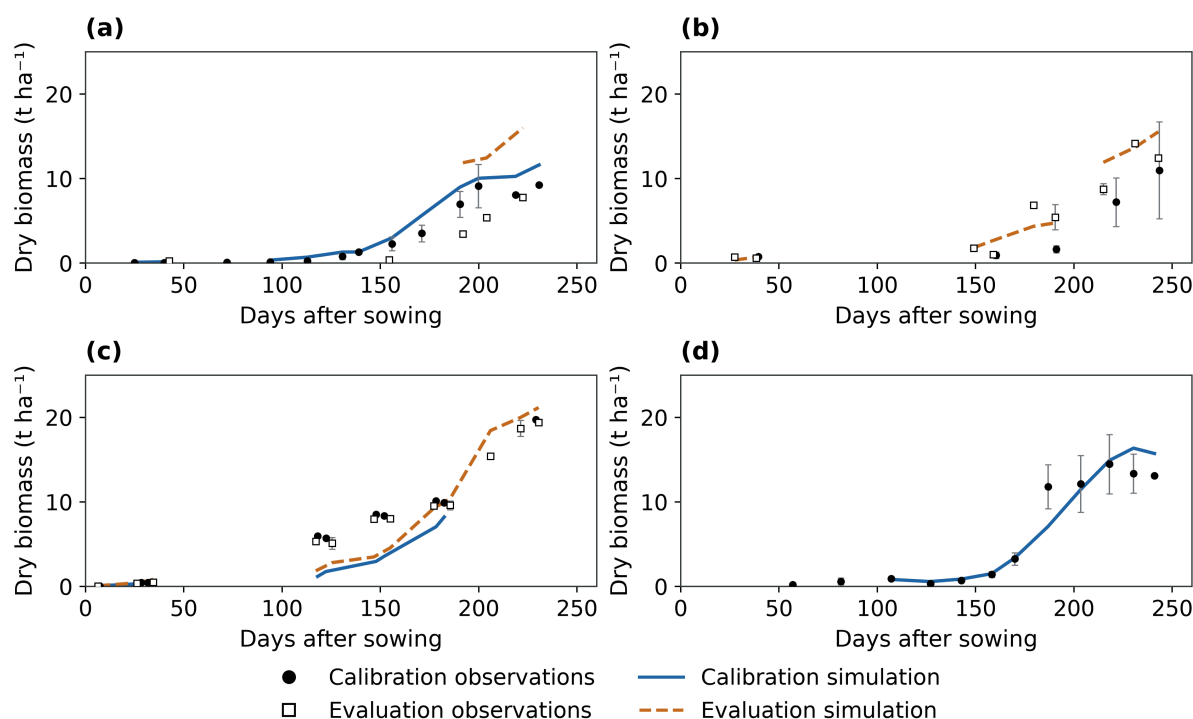
260 The original station calibration partitions retain earlier growth-fitting history; all station comparisons are
261 conditional on unconfirmed complete irrigation management. Station observations have no weight in the latest
262 water fit. Yucheng has no eligible in-season biomass evaluation observations, while its harvest-biomass endpoints
263 remain separately available. Figure S5 shows dry-grain and aboveground harvest means with quadrat SE. Six
264 samples within a crop-year describe sampling variation, and the duplicated 2012 and 2017 maize groups remain
265 excluded. Exact case availability, site-specific metrics, bin contributions and harvest source rows accompany the
266 figures.



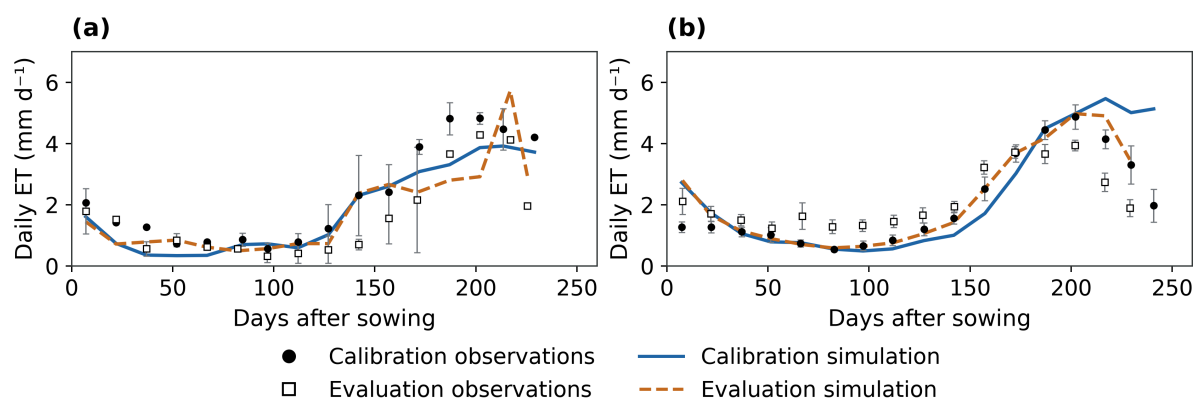
267 **Figure S5. Conditional Yucheng harvest comparisons. Wheat occupies the left column and maize the right; the top row shows**
 268 **aboveground biomass and the bottom row grain. Black points show observed means $\pm$ standard errors among six quadrat samples**
 269 **per crop-year; blue points show current coupled-model predictions at the same sampling windows. Circles and squares distinguish**
 270 **the original calibration and retrospective-evaluation partitions. Grain and total aboveground biomass are dry matter. The duplicated**
 271 **maize 2012 and 2017 source vectors remain excluded. Within-year replicates describe spatial sampling variation and are not**
 272 **independent crop seasons. Management completeness remains unconfirmed and current water coefficients use Wuqiao fitting targets,**
 273 **while growth coefficients retain their earlier multisite fitting history.**



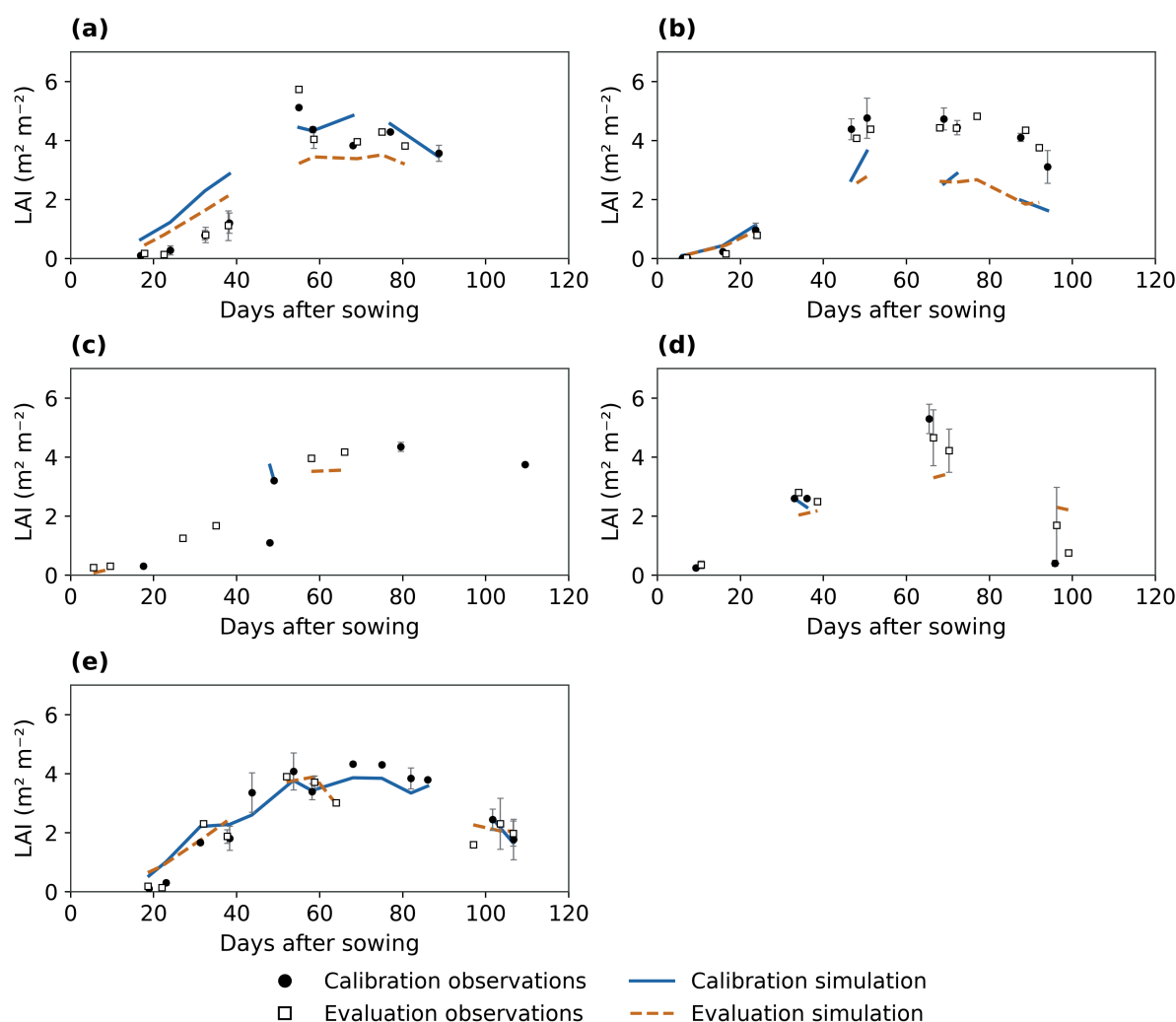
274 **Figure S6. Site-specific wheat LAI comparisons: (a) Fengqiu; (b) Luancheng; (c) Shangqiu; (d) Yucheng. Calibration and**
 275 **retrospective-evaluation observations and simulations share the same axes within each site. Error bars describe site-year**
 276 **errors; gaps are retained. The observations, weighting and conditional management interpretation follow Figure 4.**



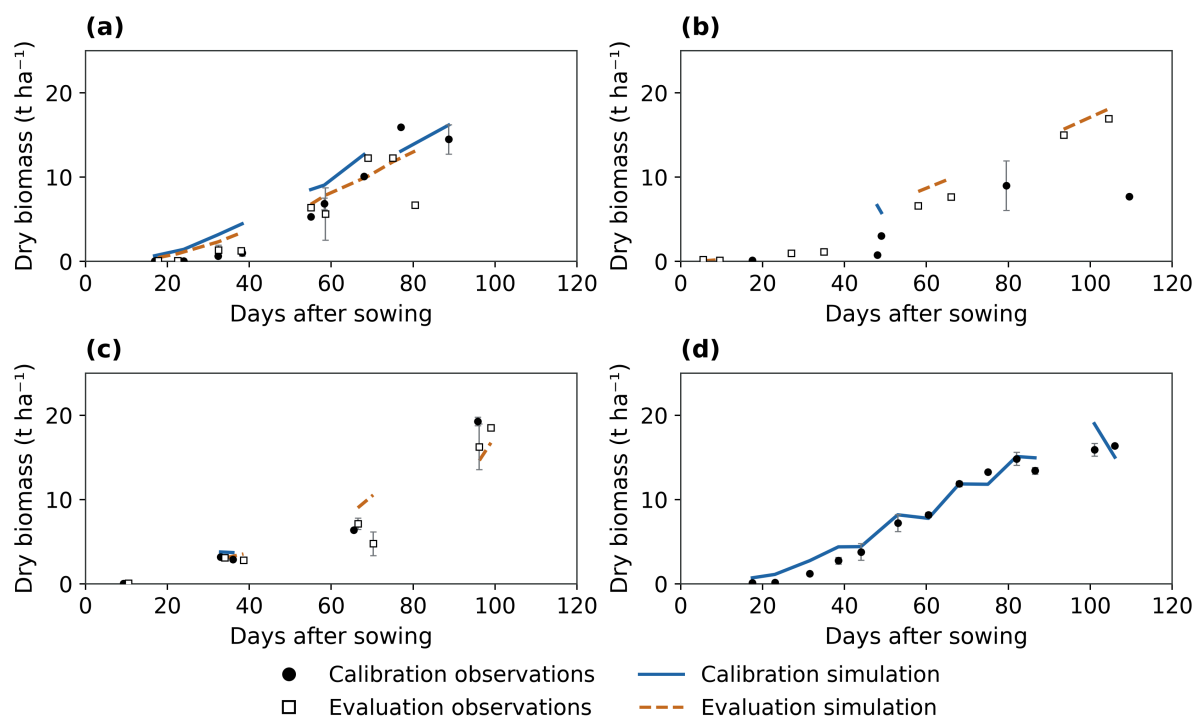
277 **Figure S7. Site-specific wheat aboveground biomass comparisons: (a) Fengqiu; (b) Luancheng; (c) Shangqiu; (d) Yucheng.**
 278 **Calibration and retrospective-evaluation observations and simulations share the same axes within each site. Error bars describe site-**
 279 **year standard errors; gaps are retained. The observations, weighting and conditional management interpretation follow Figure 4.**



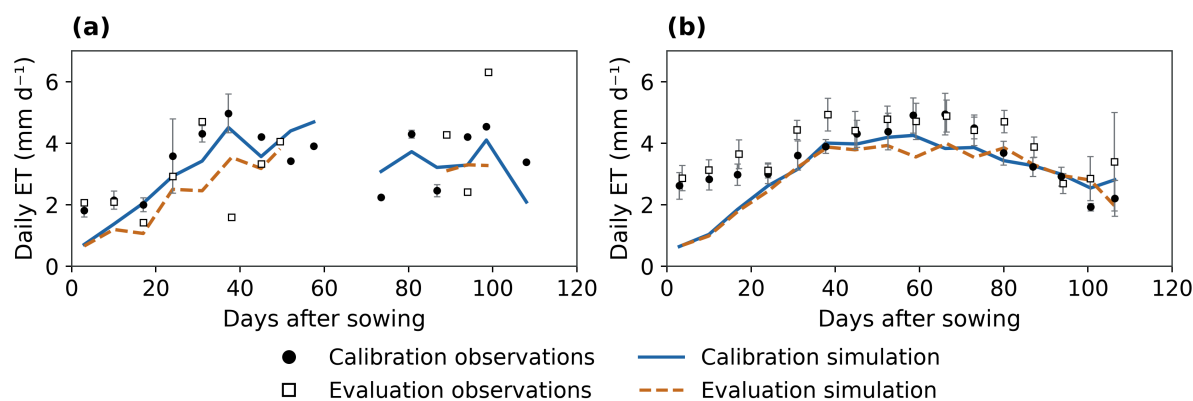
280 **Figure S8. Site-specific wheat ET comparisons: (a) Fengqiu; (b) Yucheng. Calibration and retrospective-evaluation observations and**
 281 **simulations share the same axes within each site. Error bars describe site-year standard errors; gaps are retained. The observations,**
 282 **weighting and conditional management interpretation follow Figure 4.**



283 **Figure S9. Site-specific maize LAI comparisons: (a) Fengqiu; (b) Gucheng; (c) Luancheng; (d) Shangqiu; (e) Yucheng. Calibration**
 284 **and retrospective-evaluation observations and simulations share the same axes within each site. Error bars describe site-year**
 285 **standard errors; gaps are retained. The observations, weighting and conditional management interpretation follow Figure 4.**



286 **Figure S10. Site-specific maize aboveground biomass comparisons: (a) Fengqiu; (b) Luancheng; (c) Shangqiu; (d) Yucheng.**
 287 **Calibration and retrospective-evaluation observations and simulations share the same axes within each site. Error bars describe site-**
 288 **year standard errors; gaps are retained. The observations, weighting and conditional management interpretation follow Figure 4.**



289 **Figure S11. Site-specific maize ET comparisons: (a) Fengqiu; (b) Yucheng. Calibration and retrospective-evaluation observations and**
 290 **simulations share the same axes within each site. Error bars describe site-year standard errors; gaps are retained. The observations,**
 291 **weighting and conditional management interpretation follow Figure 4.**

292 **References**

- 293 Yang, L., Fang, X., Zhou, J., Zhao, J., Hou, X., Yang, Y., Zang, H., Zeng, Z., 2024. Optimal irrigation for wheat-
 294 maize rotation depending on precipitation in the North China Plain: Evidence from a four-year experiment. Agric.
 295 Water Manag. 294, 108726. <https://doi.org/10.1016/j.agwat.2024.108726>

296 Meier, U. (Ed.), 2018. Growth stages of mono- and dicotyledonous plants: BBCH Monograph. Julius Kühn-Institut,
 297 Quedlinburg. <https://doi.org/10.5073/20180906-074619>

298 Kiniry, J.R., Jones, C.A., O'toole, J.C., Blanchet, R., Cabelguenne, M., Spanel, D.A., 1989. Radiation-use efficiency
 299 in biomass accumulation prior to grain-filling for five grain-crop species. *Field Crops Research*.
 300 [https://doi.org/10.1016/0378-4290\(89\)90023-3](https://doi.org/10.1016/0378-4290(89)90023-3)

301 de Wit; Berghuijs, n.d.. Python Crop Simulation Environment: WOFOST leaf dynamics implementation [software].
 302 Version Archived master-branch source snapshot.
 303 https://github.com/ajwdewit/pcse/blob/master/pcse/crop/leaf_dynamics.py (accessed 4 October 2026).

304 Tan, M., Gou, F., Stomph, T.J., Wang, J., Yin, W., Zhang, L., Chai, Q., van der Werf, W., 2020. Dynamic process-
 305 based modelling of crop growth and competitive water extraction in relay strip intercropping: Model
 306 development and application to wheat-maize intercropping. *Field Crops Research* 246, 107613.
 307 <https://doi.org/10.1016/j.fcr.2019.107613>

308 de Wit, A., 2020. Parameter sets for the WOFOST 7.2/7.3 cropping system model for wheat.

309 de Wit, A., 2022. Parameter sets for the WOFOST 7.2/7.3 model for simulating maize.

310 APSIM Initiative, 2026. APSIM Next Generation (ApsimX): wheat and maize model definitions [software].
 311 Version 074d15f.
 312 <https://github.com/APSIMInitiative/ApsimX/tree/074d15f51889a386d3667acfed184d03e4e8f65f/Models/Resources>
 313 rces (accessed 4 October 2026).

314 Allen, R.G., Pereira, L.S., Raes, D., Smith, M., 1998. Crop evapotranspiration: Guidelines for computing crop
 315 water requirements, FAO Irrigation and Drainage Paper. Food and Agriculture Organization of the United
 316 Nations, Rome.

317 van Genuchten, M.T., 1980. A Closed-form Equation for Predicting the Hydraulic Conductivity of Unsaturated
 318 Soils. *Soil Science Society of America Journal* 44, 892–898.
 319 <https://doi.org/10.2136/sssaj1980.03615995004400050002x>

320 Mualem, Y., 1976. A new model for predicting the hydraulic conductivity of unsaturated porous media. *Water*
 321 *Resources Research* 12, 513–522. <https://doi.org/10.1029/wr012i003p00513>

- 322 Wu, L., Min, L., Liu, M., Zhang, Y., Pei, H., Li, H., Zhang, G., Wang, S., Shen, Y., 2023. Monitoring of Thick
323 Vadose Zone Water Dynamics Under Irrigation Using a 48 m Deep Caisson at the Luancheng Critical Zone
324 Observatory. *Water Resour. Res.* 59, e2022WR032965. <https://doi.org/10.1029/2022wr032965>
- 325 Saxton, K.E., Rawls, W.J., 2006. Soil Water Characteristic Estimates by Texture and Organic Matter for Hydrologic
326 Solutions. *Soil Science Society of America Journal* 70, 1569–1578. <https://doi.org/10.2136/sssaj2005.0117>
- 327 ISRIC — World Soil Information, 2020. SoilGrids250m version 2.0.
- 328 NASA JPL, 2013. NASA Shuttle Radar Topography Mission Global 1 arc second.
329 <https://doi.org/10.5067/MEASURES/SRTM/SRTMGL1.003>
- 330 Qiu, B., Hu, X., Chen, C., Tang, Z., Yang, P., Zhu, X., Yan, C., Jian, Z., 2022. Maps of cropping patterns in China
331 during 2015–2021. *Scientific Data*. <https://doi.org/10.1038/s41597-022-01589-8>
- 332 Fu, Y., Dong, J., Shen, R., Peng, Q., Li, X., Zheng, Y., Pan, B., Dong, J., Qiu, B., Huang, J., Yuan, W., 2025. China
333 Crop Dataset: A long-term and high-resolution crop distribution dataset in China.
334 <https://doi.org/10.57760/sciencedb.32361>
- 335 Fu, Y., Dong, J., Shen, R., Peng, Q., Li, X., Zheng, Y., Pan, B., Dong, J., Qiu, B., Huang, J., Huang, X., Yuan, W.,
336 2026. A 30-m annual distribution dataset of major crops in China from 2001-2024. *Scientific Data* 13, 1074.
337 <https://doi.org/10.1038/s41597-026-07370-5>